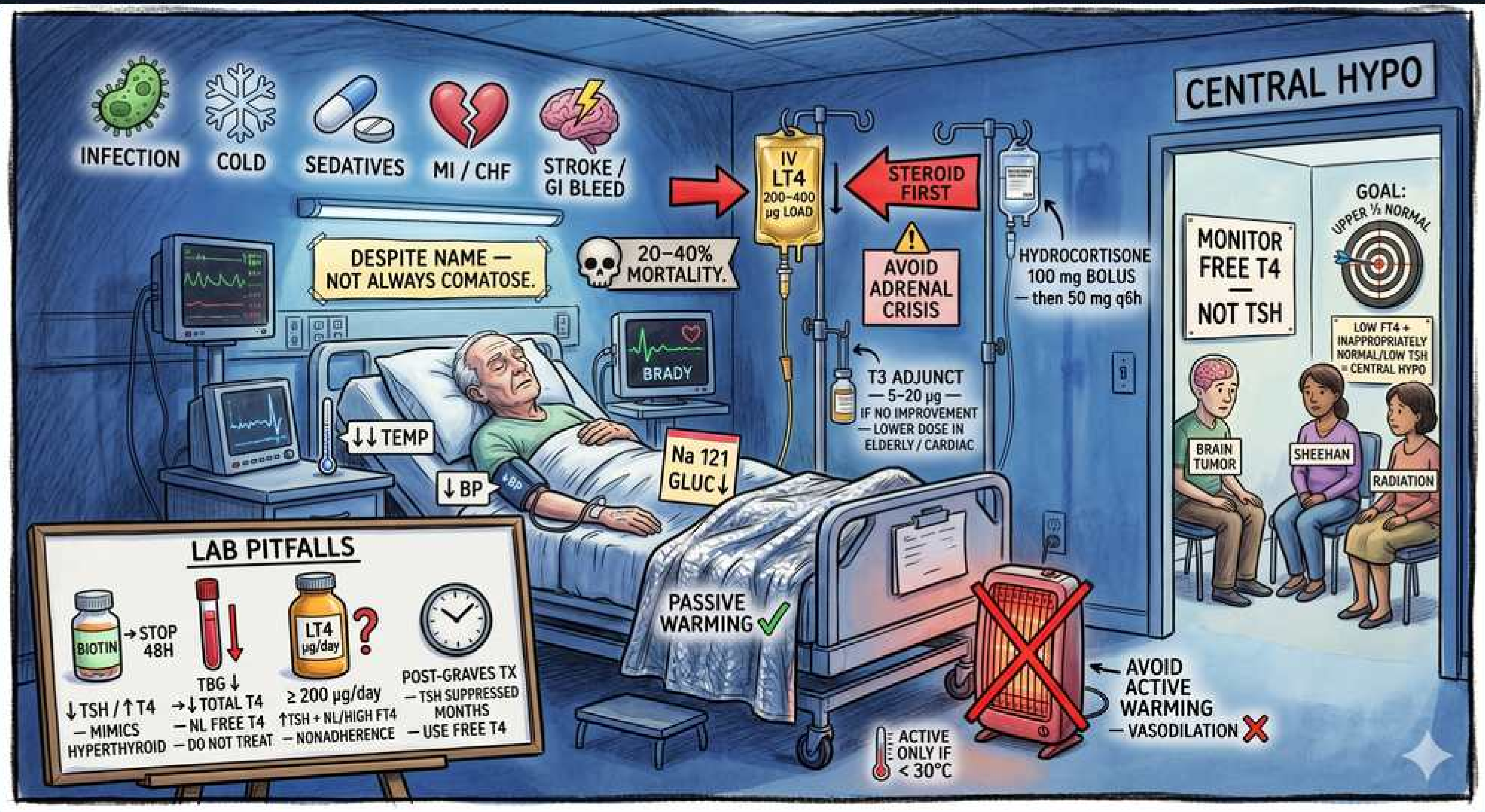
Thyroid — Myxedema Coma & Central Hypothyroidism



1. Precipitants

WHAT YOU SEE

Infection, cold, sedatives, MI/CHF, stroke/GI bleed icons

WHAT IT ENCODES

Myxedema coma is almost always decompensated chronic hypothyroidism unmasked by an acute stressor — infection/sepsis (most common), cold exposure (classic winter presentation in elderly women), sedatives/opioids/anesthesia, MI or CHF, stroke, and GI bleed. Non-adherence to levothyroxine is a frequent under-recognized trigger. Mnemonic: hypothyroid patient + a stressor + hypothermia + ↓consciousness = think myxedema coma and treat empirically.

BOARD TRAP

Waiting for confirmatory TSH/free-T4 before treating — labs take time and the diagnosis is clinical; empiric therapy must start on suspicion.

---

2. High mortality

WHAT YOU SEE

Skull '20–40% MORTALITY'

WHAT IT ENCODES

Even with treatment, mortality is 20–40%, driven by the underlying cardiorespiratory failure, hypothermia, and the precipitant rather than the hypothyroidism alone. This is an ICU-level endocrine emergency. Poor prognostic markers: persistent hypothermia, hypotension unresponsive to pressors, bradycardia, low GCS, and high APACHE II.

---

3. Not always comatose

WHAT YOU SEE

'DESPITE NAME — NOT ALWAYS COMATOSE'

WHAT IT ENCODES

The name is a misnomer — most patients are obtunded/confused, not frankly comatose. The syndrome is decompensated hypothyroidism: altered mentation, hypothermia, hypoventilation with CO2 retention, bradycardia, and hypotension. 'Myxedema' refers to the non-pitting mucopolysaccharide skin/soft-tissue infiltration (puffy face, macroglossia, periorbital edema).

BOARD TRAP

Excluding the diagnosis because the patient is awake — altered mentation without coma still qualifies, and delaying treatment is fatal.

---

4. IV levothyroxine

WHAT YOU SEE

'IV LT4 200–400 µg LOAD'

WHAT IT ENCODES

Treat with an IV levothyroxine loading dose of 200–400 µg (lower end for small/elderly/cardiac patients), then 50–100 µg IV daily. IV is preferred because gut absorption is unreliable in a hypothermic, hypoperfused patient. T4 has a long half-life (~7 days) so a load rapidly replenishes the depleted pool.

BOARD TRAP

Oral-only dosing in a patient with gut hypomotility/edema — absorption is too slow and unpredictable in true myxedema coma.

---

5. Steroids FIRST

WHAT YOU SEE

A red arrow 'STEROID FIRST' pointing ahead of the LT4 bag, captioned 'AVOID ADRENAL CRISIS' — steroids must go in before thyroid hormone or you trigger adrenal crisis.

WHAT IT ENCODES

Give IV hydrocortisone BEFORE (or simultaneously with) thyroid hormone. Coexisting adrenal insufficiency is common (autoimmune polyglandular disease, or central/pituitary causes affecting both axes). Thyroid hormone increases metabolic clearance of cortisol — giving T4 first to an undiagnosed adrenal-insufficient patient can precipitate fatal adrenal crisis.

BOARD TRAP

Giving thyroid hormone before steroids — the single most-tested sequencing pitfall; it can trigger adrenal crisis.

---

6. Hydrocortisone dosing

WHAT YOU SEE

'HYDROCORTISONE 100mg BOLUS then 50 q6h'

WHAT IT ENCODES

Empiric stress-dose hydrocortisone: 100 mg IV bolus then 50 mg IV q6–8h (≈200–300 mg/day) until adrenal insufficiency is excluded (ideally draw a random/stimulated cortisol first if it won't delay care). Hydrocortisone is chosen because it covers both glucocorticoid and mineralocorticoid needs.

---

7. T3 adjunct cautious

WHAT YOU SEE

'T3 ADJUNCT 5–20µg, LOWER DOSE IN ELDERLY/CARDIAC'

WHAT IT ENCODES

T3 (liothyronine) 5–20 µg IV may be added because peripheral T4→T3 conversion is impaired in critical illness, giving faster onset. However T3 raises the risk of arrhythmia and ischemia, so use lower doses (or avoid) in elderly and cardiac patients. Many protocols give T4 alone ± low-dose T3 if no improvement.

BOARD TRAP

Aggressive high-dose T3/T4 in an elderly cardiac patient → MI, arrhythmia, or death from a sudden metabolic surge.

---

8. Hypothermia / passive warming

WHAT YOU SEE

A green-checked 'PASSIVE WARMING' (blankets) beside a space heater stamped with a big red X 'AVOID ACTIVE WARMING — VASODILATION', and a thermometer 'ACTIVE ONLY IF <30°C' — passive blankets are right; active heat causes vasodilatory collapse.

WHAT IT ENCODES

Hypothermia is a defining feature. Use PASSIVE external rewarming (blankets) — aggressive active rewarming causes peripheral vasodilation that worsens hypotension and can cause cardiovascular collapse. Reserve active warming for severe hypothermia (<30°C). Core temperature corrects mainly as thyroid hormone is replaced.

BOARD TRAP

Aggressive active rewarming (warming blankets/baths) → peripheral vasodilation → vasodilatory shock and collapse.

---

9. Hyponatremia & hypoglycemia

WHAT YOU SEE

'Na 121, GLUC↓'

WHAT IT ENCODES

Hyponatremia (e.g., Na 121) results from impaired free-water excretion (SIADH-like, reduced renal perfusion); correct slowly with fluid restriction ± hypertonic saline only if seizing — avoid overcorrection. Hypoglycemia reflects adrenal insufficiency and hypometabolism; give IV dextrose. Also expect hypoventilation/hypercapnia (may need ventilation) and bradycardia.

BOARD TRAP

Rapid sodium correction → osmotic demyelination; or attributing hyponatremia to one cause without addressing thyroid/adrenal axes.

---

10. Central hypothyroidism causes

WHAT YOU SEE

'BRAIN TUMOR, SHEEHAN, RADIATION'

WHAT IT ENCODES

Central (secondary/tertiary) hypothyroidism arises from pituitary or hypothalamic disease: pituitary tumor/adenoma, Sheehan's syndrome (postpartum pituitary infarction), cranial/pituitary radiation, infiltrative disease (sarcoid, hemochromatosis), and trauma. Suspect when multiple pituitary axes are affected — always screen for and treat coexisting adrenal insufficiency first.

BOARD TRAP

Treating central hypothyroidism with T4 before checking the adrenal axis — can precipitate adrenal crisis just as in myxedema coma.

---

11. Central: monitor free T4 not TSH

WHAT YOU SEE

A bedside monitor placard 'MONITOR FREE T4 — NOT TSH' — in central disease the broken pituitary makes TSH useless, so you track free T4 instead.

WHAT IT ENCODES

In central hypothyroidism the pituitary cannot mount an appropriate TSH response, so TSH is low, normal, or only mildly elevated despite a low free T4 — TSH is uninterpretable for diagnosis and monitoring. Diagnose and titrate replacement to a free T4 in the mid-to-upper normal range, not to TSH.

BOARD TRAP

Using TSH to diagnose or titrate central hypothyroidism — a 'normal' TSH falsely reassures while the patient remains hypothyroid.

---

12. Biotin pitfall — stop 48h

WHAT YOU SEE

A supplement bottle 'BIOTIN — STOP 48h' captioned 'mimics hyperthyroid (↓TSH ↑T4)' — biotin fools the assay into a fake Graves' pattern, so stop it before testing.

WHAT IT ENCODES

High-dose biotin (supplements, hair/nail products) interferes with streptavidin-biotin immunoassays, producing a falsely LOW TSH and falsely HIGH free T4/T3 — a pattern that mimics Graves'/hyperthyroidism (and can also distort thyroid-antibody and troponin assays). Stop biotin ~48–72h before testing and repeat.

BOARD TRAP

Treating a biotin-induced 'hyperthyroid' lab pattern as real disease — a clinically euthyroid patient with this discordant panel should prompt a biotin history and a repeat off biotin.

---

13. Sick euthyroid (TBG/T4 traps)

WHAT YOU SEE

A LAB PITFALLS panel '↓TBG ↓TOTAL T4, NL FREE T4 → DO NOT TREAT' — total T4 looks low but the normal FREE T4 means it is sick-euthyroid, not real hypothyroidism.

WHAT IT ENCODES

Non-thyroidal illness (sick euthyroid) in hospitalized/critically ill patients: low T3 first (↓peripheral T4→T3 conversion, ↑reverse T3), then low total T4 — but free T4 and TSH are typically normal and the patient is not truly hypothyroid. Low TBG (e.g., nephrotic/cirrhosis, androgens) also lowers TOTAL T4 with a normal FREE T4. Do not replace; recheck after recovery.

BOARD TRAP

Treating sick-euthyroid lab changes with levothyroxine — replacement does not help and may harm; avoid checking thyroid panels during acute illness unless myxedema coma is suspected.

---

14. Post-Graves use T4

WHAT YOU SEE

'POST-GRAVES TX MONTHS, TSH SUPPRESSED — USE FREE T4'

WHAT IT ENCODES

After definitive Graves' treatment (RAI, surgery, or thionamides), the pituitary thyrotrophs stay suppressed and TSH can remain low for weeks to months even once the patient is euthyroid or becoming hypothyroid. During this lag, titrate replacement to FREE T4, not the still-suppressed TSH.

BOARD TRAP

Over-replacing (or under-treating) because TSH is still suppressed post-Graves — the lag is physiologic; follow free T4 until TSH recovers.
